

Day 18 Notes – AWS S3 (Simple Storage Service)

1. What is Amazon S3?

Amazon S3 (Simple Storage Service) is an object storage service used to store and retrieve any amount of data from anywhere on the internet.

Features

- Highly Scalable
 - Highly Available
 - Highly Durable
 - Secure
 - Cost Effective
 - Event Driven
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2. S3 Components

Bucket

- Container used to store objects.
- Bucket names must be globally unique.

Example:

sai-cloud-training-bucket

Object

An object consists of:

- Data (File)
- Metadata
- Key (File Name)

Example:

image.jpg

document.pdf

backup.zip

3. S3 Storage Classes

S3 Standard

- Frequently accessed data
- Multi-AZ storage
- Low latency

Use Cases:

- Websites
- Mobile applications
- Dynamic content

S3 Intelligent-Tiering

- Automatically moves data between tiers based on access patterns.
- Cost optimization.

Use Cases:

- Unknown access patterns

S3 Standard-IA

(IA = Infrequent Access)

- Less frequently accessed data
- Lower storage cost
- Retrieval charges apply

Use Cases:

- Backups
- Disaster Recovery

S3 One Zone-IA

- Stored in a single AZ
- Lower cost than Standard-IA

Use Cases:

- Secondary backups
- Re-creatable data

S3 Glacier Instant Retrieval

- Rarely accessed data
- Millisecond retrieval

Use Cases:

- Medical records
- Archives

S3 Glacier Flexible Retrieval

- Archive storage
- Retrieval in minutes to hours

Use Cases:

- Long-term backups

S3 Glacier Deep Archive

- Lowest-cost storage
- Retrieval may take hours

Use Cases:

- Compliance records
- Historical data

4. S3 Availability vs Durability

Availability

Measures how often data can be accessed.

Example:

Availability = 99.99%

If S3 has 99.99% availability:

Downtime per year $\approx$ 52 minutes

Durability

Measures data protection against loss.

S3 Durability:

99.999999999%

(11 Nines Durability)

Meaning:

If you store:

10,000,000 objects

You may lose:

1 object every 10,000 years

5. Availability and Durability Comparison

Storage Class	Availability Durability	
Standard	99.99%	99.999999999%
Intelligent-Tiering	99.9%+	99.999999999%
Standard-IA	99.9%	99.999999999%
One Zone-IA	99.5%	99.999999999%

6. S3 Versioning

Purpose:

- Protect against accidental deletion
- Recover previous versions

Example:

file.txt (v1)
file.txt (v2)
file.txt (v3)

7. S3 Lifecycle Policies

Automatically move data between storage classes.

Example:

Day 0 -> Standard
Day 30 -> Standard-IA
Day 90 -> Glacier
Day 365 -> Delete

Benefits:

- Cost Optimization
 - Automated Data Management
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8. S3 Security

Bucket Policies

Resource-based permissions.

IAM Policies

User/Role permissions.

ACL (Access Control List)

Legacy permission model.

Block Public Access

Recommended for all buckets.

Encryption

SSE-S3

AWS Managed Keys

SSE-KMS

Customer-managed keys via KMS

SSE-C

Customer provides encryption key

9. S3 Static Website Hosting

Steps:

1. Create Bucket
2. Upload HTML files
3. Enable Static Website Hosting
4. Configure Bucket Policy
5. Access Website URL

Example:

index.html

error.html

10. S3 Event Notifications

Triggers events when:

- Object Created
- Object Deleted
- Object Restored

Targets:

- Lambda
- SNS
- SQS

Example:

Upload Image



Lambda Trigger



Thumbnail Creation

11. S3 Replication

Cross Region Replication (CRR)

Replicate data between regions.

Example:

Mumbai → Singapore

Same Region Replication (SRR)

Replicate within the same AWS region.

Interview Questions

1. What is the difference between S3 Availability and Durability?
2. What are the different S3 Storage Classes?
3. What is Versioning in S3?
4. What is Lifecycle Management?
5. Difference between CRR and SRR?
6. What is S3 Intelligent-Tiering?
7. How do you secure an S3 Bucket?
8. What is Static Website Hosting?
9. What is an S3 Event Notification?
10. Why is S3 called Object Storage?